adapted on
Lynford L. Goddard
Brian T. Cunningham

ECE 329

Fields and Waves I

Prof. Victoria Shao
Teaching Associate Professor

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Prof. Victoria Shao

Email: yangshao@Illinois.edu
Office: ECEB 5048

Class email for communication:
ece-329-fall24-group@office365.illinois.edu

Research interests

- Electromagnetic compatibility/interference (EMC/EMI)
  - ECE 498 Engineering EMC – Prof. Victoria Shao
- Signal and power integrity analysis (SI/PI)
  - ECE 546 Advanced Signal Integrity – Prof. Jose Schutt-Aine
- Computational electromagnetics (CEM)
  - ECE 540 Computational Electromagnetics – Prof. Jianming Jin
- High-power microwave (HPM)

Outside Classroom

- Two kids (10th grade and 1st grade)
- Playing Volleyball
- Watch Sci Fi movies and criminal TV series
- Have a dog named Molly





Course Organization

- Section X: MWF 12:00pm-12:50pm
- Course website
  - https://courses.engr.illinois.edu/ece329/index.html
  - Syllabus
  - Homework due **Thursday** at **6pm**
  - Grading policy
  - Canvas, discussion forum, Homework Solutions, Slides
  - Homework assignment on Gradescope
- 3 mid-terms and 1 final
- Office hour: Tuesday 10am-11pm
- Prerequisites: MATH 241, PHYS 212, ECE 210
- Follow up course: ECE 350

Course Organization

- Grade breakdown

Homework	15% of total
Midterm Exams (x3)	60% of total
Final Exam	25% of total

- Midterm time

Exam I:	Monday- 7:00pm-8:15pm, February 19, 2024. ECEB 1002(Grainger Auditorium)
	Conflicts: Tuesday- 8:15am-9:30am, February 20, 2024. Location TBD.
Exam II:	Monday- 7:00pm-8:15pm, March 25, 2024. ECEB 1002(Grainger Auditorium)
	Conflicts: Tuesday- 8:15am-9:30am, March 26, 2024. Location TBD.
Exam III:	Monday- 7:00pm-8:15pm, April 22, 2024. ECEB 1002(Grainger Auditorium)
	Conflicts: Tuesday- 8:15am-9:30am, April 23, 2024. Location TBD.

How to get an A in ECE329

- ***Time Management.*** Allocate 10 hrs/wk of *regularly scheduled times* in the week outside of class for 329:
  - 30min for **reading** of textbook **before** each class
  - 30min for **studying** online notes **before** each class
  - 30min for **studying** my notes **between** classes
  - 5-6 hrs/wk for HWs
    - In a semester, all lectures total only 32.5hrs, which is less than 1 week at a job! It's up to you to put in the time to learn
- Get a 1" binder (organize lecture notes/HWs/exams)
- Start assignments early. **Do all problems by yourself first.** If you get stuck, form study groups to work on problems together but **ALWAYS** write-up and submit **YOUR OWN** solutions. Do not blindly copy.
- Ask questions and come to office hours if you get stuck. Don't let confusion snowball

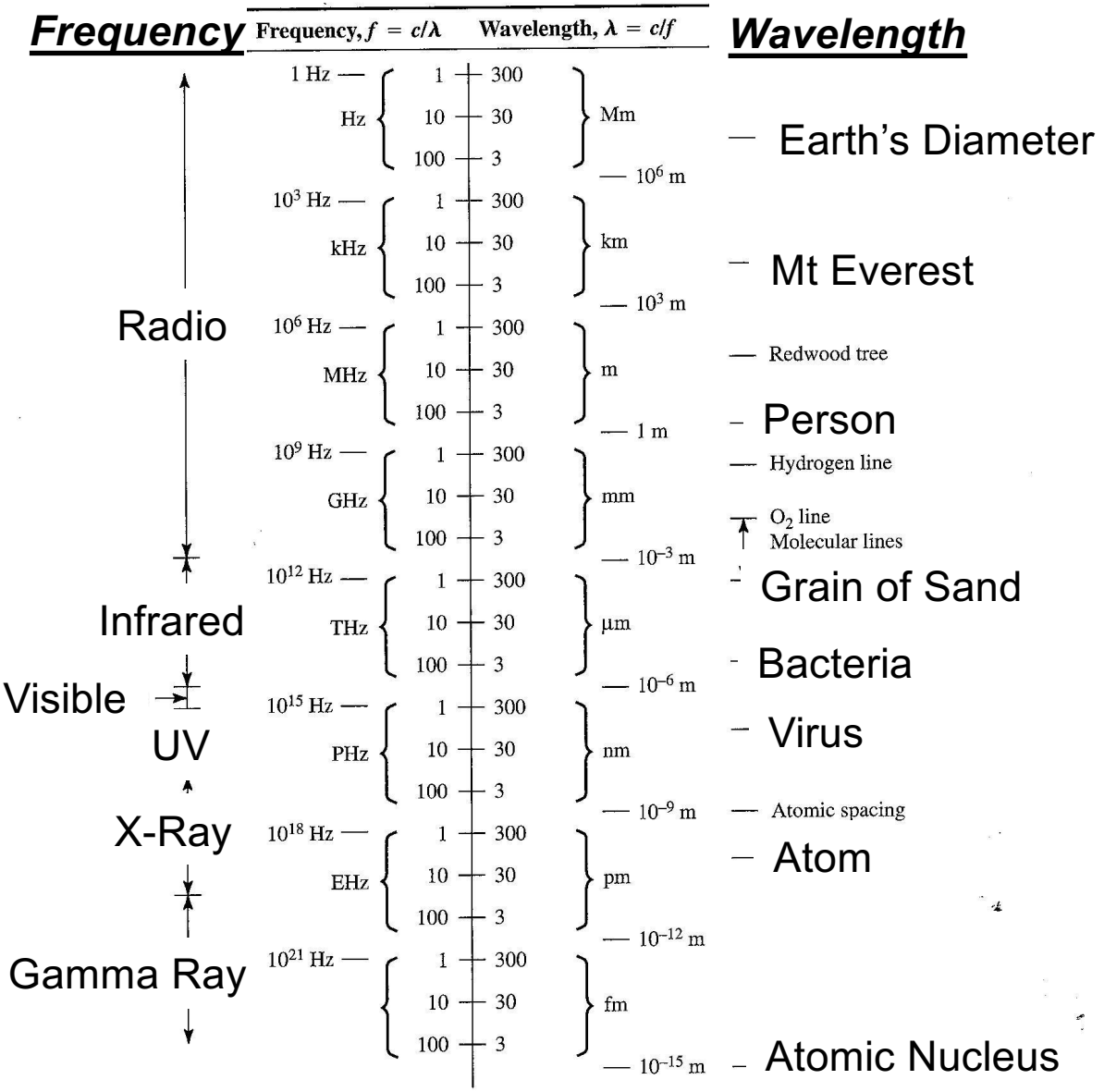
How to get an A in ECE329

- ***Practice doing problems.*** Get comfortable with the math manipulations and associated physical meaning, and you will find exam problems to be easier
  - HW problems
  - Example problems worked in lecture
  - Example problems worked in online class notes
  - Old exam problems
  - Office hours
- ***Review your prerequisites.***
  - Vector calculus, line/surface integrals: Math 241
  - Linear circuit, system analysis, phasors: ECE 110/210
  - Electric and magnetic fields: Physics 212
- ***Come to class!!***
  - I will discuss topics to be emphasized on exams and give hints about how to approach the more difficult homework problems.

EM Spectrum

$\lambda f = c$

TABLE 1-1
Frequencies and wavelengths of the electromagnetic spectrum from almost dc to gamma-rays



Speed of Light

Distance Travelled	Travel Time
Across a virus	100×10^{-18} sec
300 meters	1 microsecond
NY to LA	13 milliseconds
Around earth	0.133 sec
Earth to Moon	1.2 sec
Earth to Sun	8.3 minutes
Earth to Mars	3-21 minutes
Sun to nearest star	4 years
Diameter of our galaxy	100,000 years
Edge of known universe	15 billion years

c ≡ 299,792,458 m/s (used to define the meter!)

Sending/Receiving EM Waves



Radio telescope



Cell phone tower
Satellite receiver

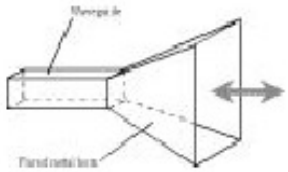


Cell phone





Television



Directional antenna



Wireless Internet



Microwave patch antenna

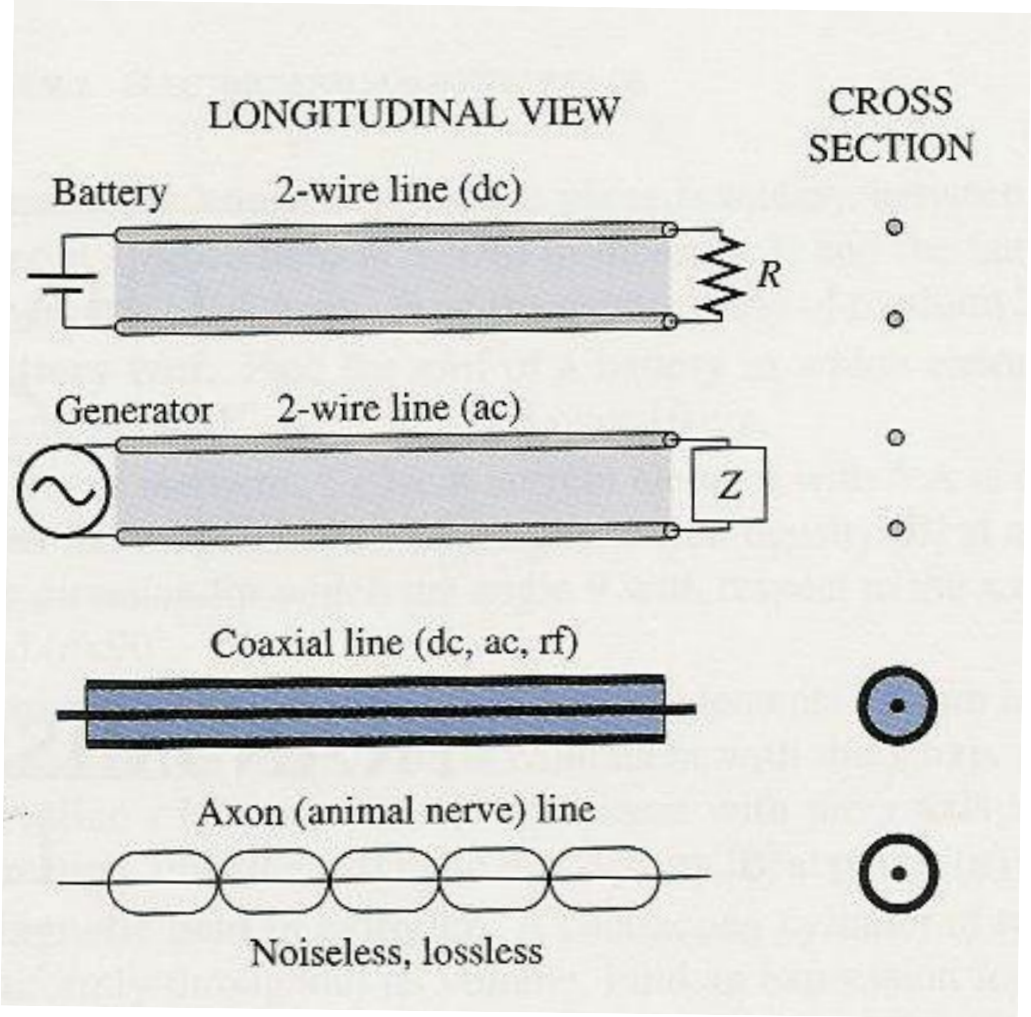


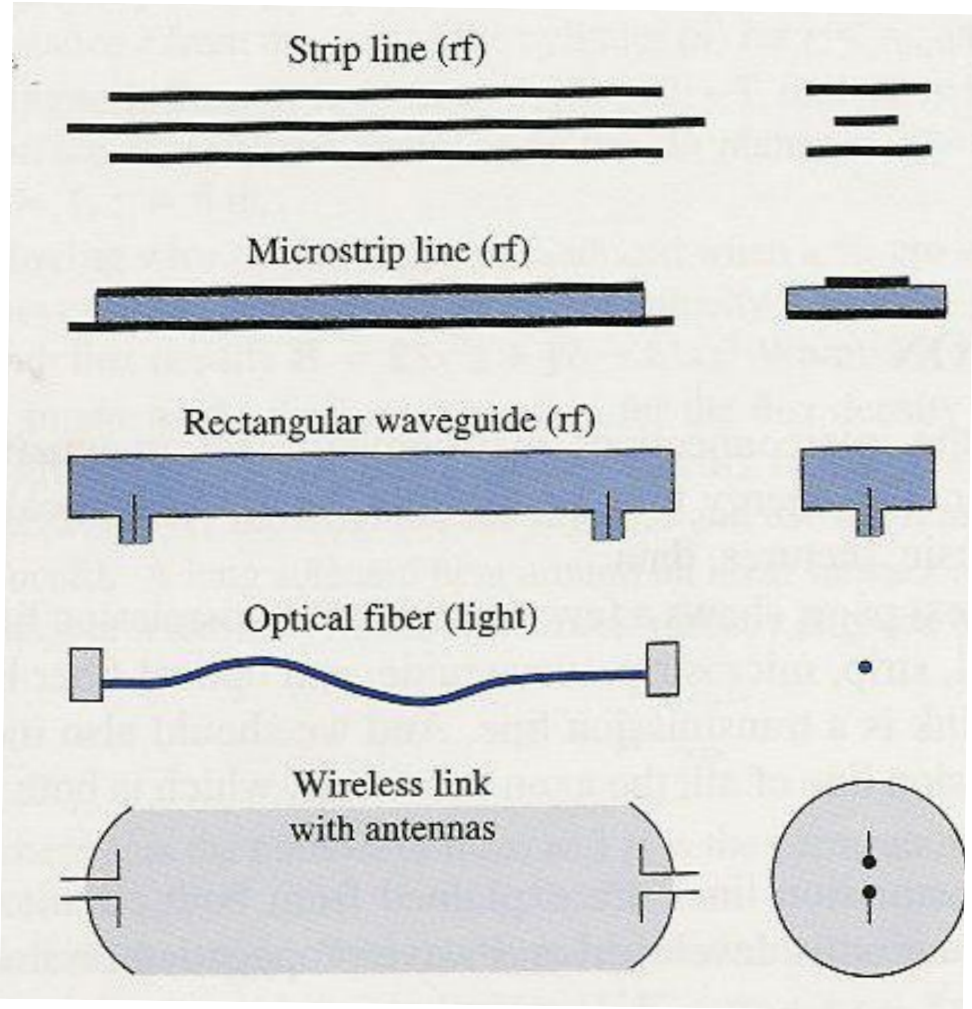
RF ID Tag



Arecibo Observatory

Guiding EM Waves



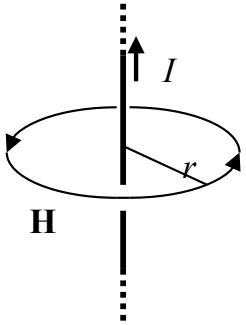


Brief History of EM

Charles Coulomb in 1785 demonstrated how electric charges repel one another



Andre Marie Ampere discovered that an electric current produces a magnetic field in 1820





Michael Faraday in 1831 showed that since an electric current could produce a magnetic field, a changing magnetic field can produce an electric current. "Principle of Induction" used for first electric generators.



James Clerk Maxwell (1831-1879)

- 1855-1868 - formulates field equations for electromagnetism. Predicts existence of EM wave propagation and the speed of light. Shows theoretical possibility of generating electromagnetic radiation
- 1873: publishes *Treatise on Electricity and Magnetism*



Maxwell's Equations

Integral form

$$\oint_C \mathbf{E}\cdot d\mathbf{L} = -\frac{d}{dt}\int_S \mathbf{B}\cdot d\mathbf{S}$$ Faraday

$$\oint_C \mathbf{H}\cdot d\mathbf{L} = \int_S \mathbf{J}\cdot d\mathbf{S} + \frac{d}{dt}\int_S \mathbf{D}\cdot d\mathbf{S}$$ Ampere

$$\oint_S \mathbf{D}\cdot d\mathbf{S} = \int_{Vol} \rho\, dv$$ Gauss

$$\oint_S \mathbf{B}\cdot d\mathbf{S} = 0$$ Gauss

Differential Form

$$\nabla\times\mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t}$$

$$\nabla\times\mathbf{H} = \mathbf{J} + \frac{\partial\mathbf{D}}{\partial t}$$

$$\nabla\cdot\mathbf{D} = \rho$$

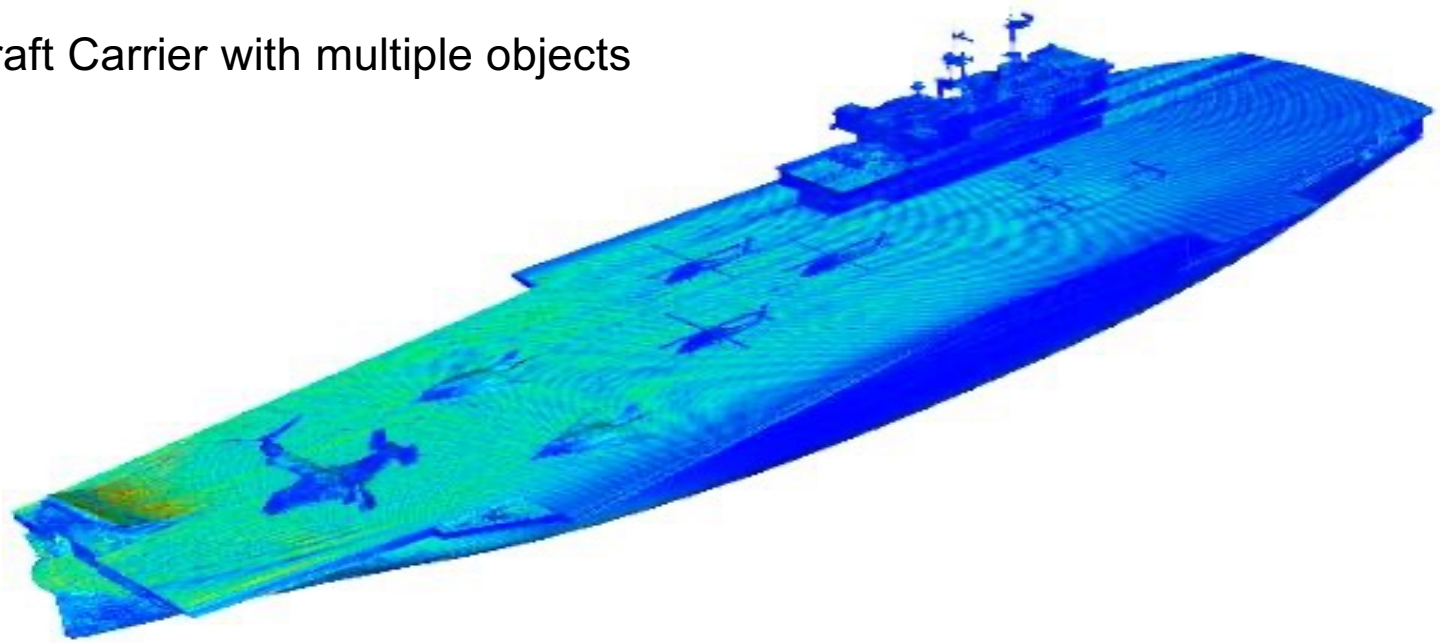
$$\nabla\cdot\mathbf{B} = 0$$

E Electric Field
H Magnetic Field
D=ε**E** "Displacement Flux Density"
B=μ**H** "Magnetic Flux Density"
J Current Density
ρ Charge Density

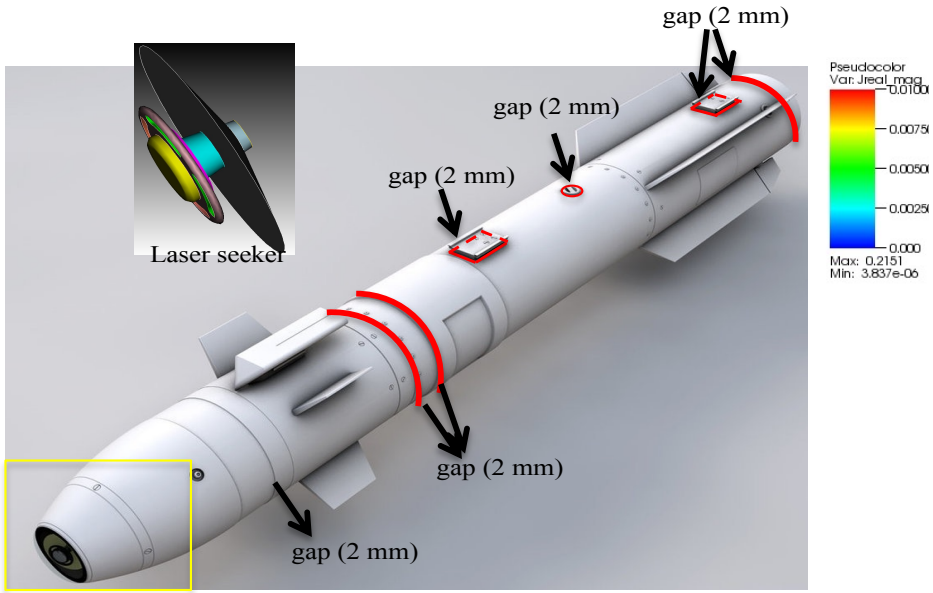
Computers use Differential Form for Complex Objects

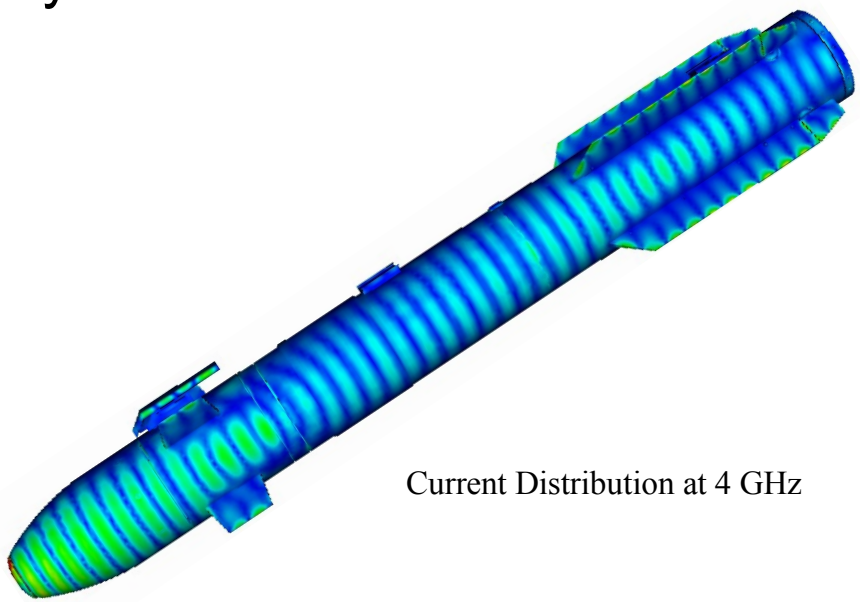
From huge objects to the nanoscale, Maxwell's Equations always work!

Aircraft Carrier with multiple objects



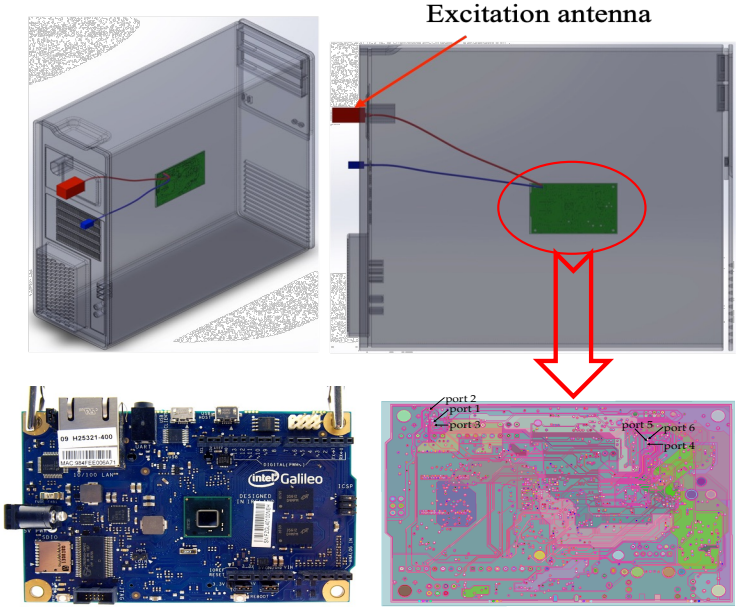
Electromagnetic Susceptibility of Hellfire Missile

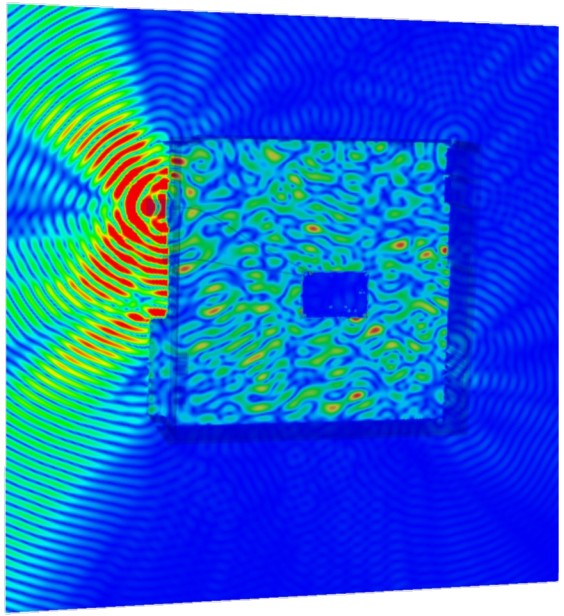




Current Distribution at 4 GHz

Circuit/Antenna Analysis





E field distribution

Units and Constants

Electric field **E**: _____

Magnetic field **H**: _____

Displacement flux density **D**: _____

Magnetic flux density **B**: _____

Current density ***J***: _____

Charge density **ρ**: _____

$$\varepsilon_o \approx 8.854\times10^{-12}\ F/m$$

$$\mu_o = 4\pi\times10^{-7}\ H/m$$

$$c \approx 3\times10^{8}\ m/s$$

Sections 1.1-1.2

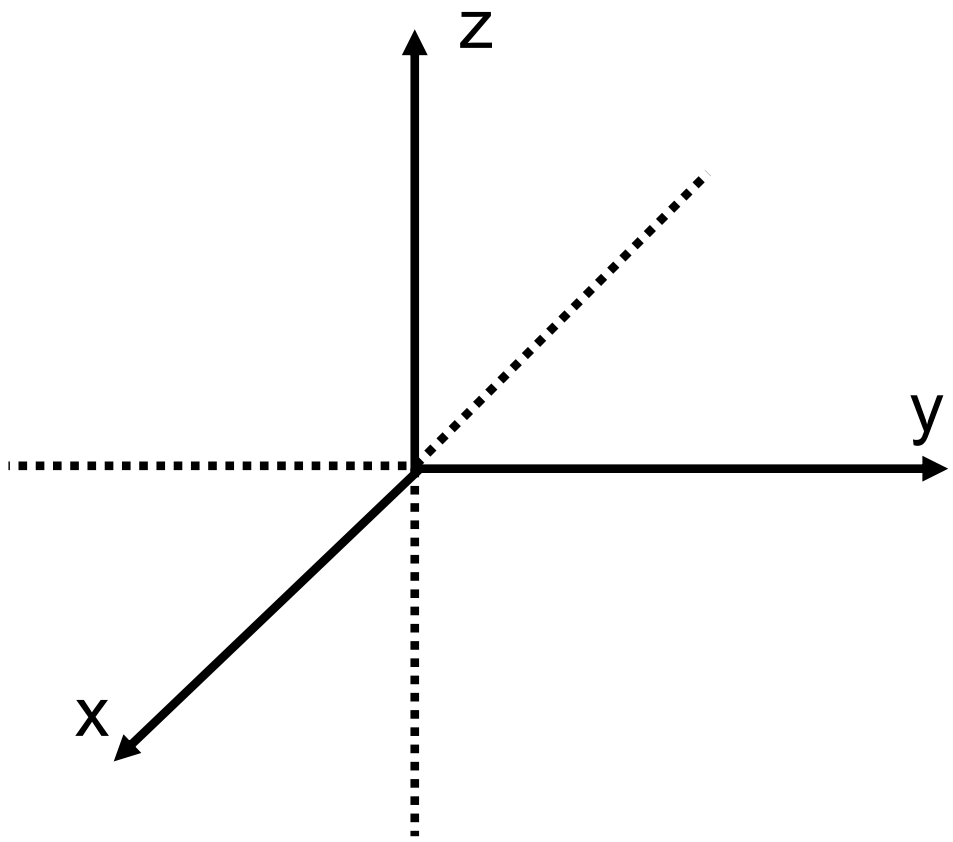
Vector Algebra
Cartesian Coordinates
Differential Length Vector
Differential Surface Vector

Adapted from Prof. Cunningham's Notes

Today's Topics

- Review
  - Scalars (numbers) and vectors
  - Unit vectors
  - Vector addition & subtraction
  - Magnitude
  - Dot product
  - Cross product
- New Topics
  - Differential length vector
  - Differential surface vector

Cartesian Coordinate System



Unit vectors

$\hat{a}_x \quad \hat{a}_y \quad \hat{a}_z$

Notation

$$\vec{A} = A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z$$

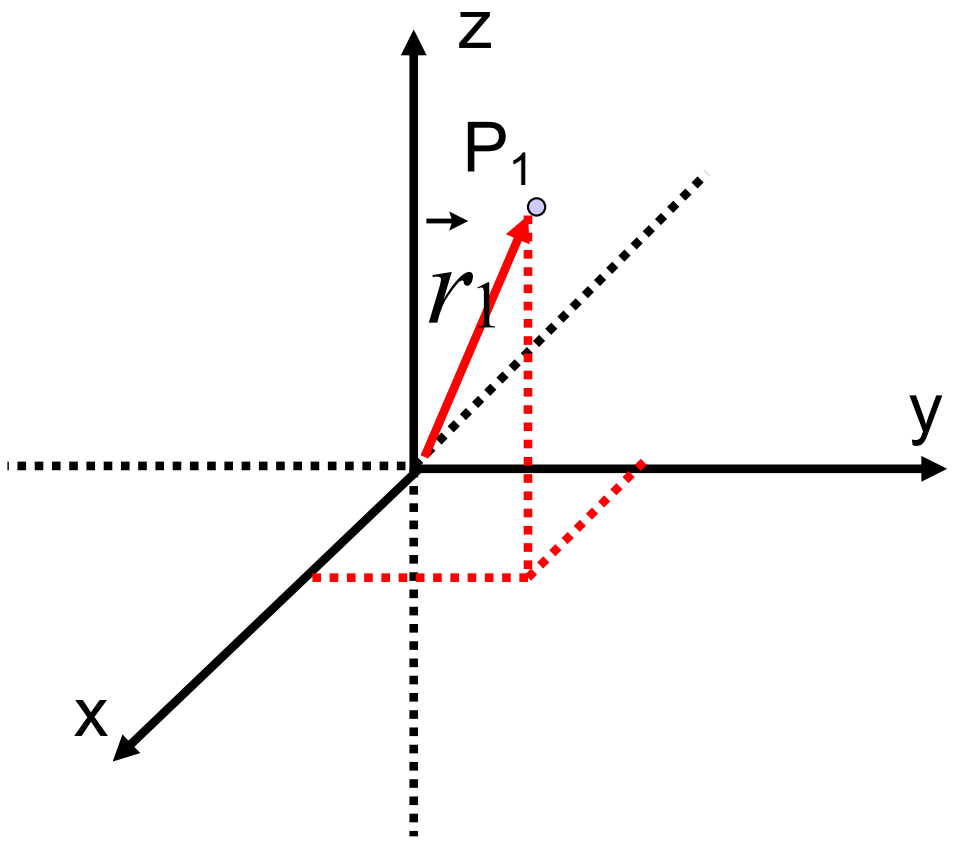
Alternate Notation

$$\vec{A} = < A_x, A_y, A_z >$$

$$\vec{A} = A_1\hat{a}_1 + A_2\hat{a}_2 + A_3\hat{a}_3$$

(Review at home)

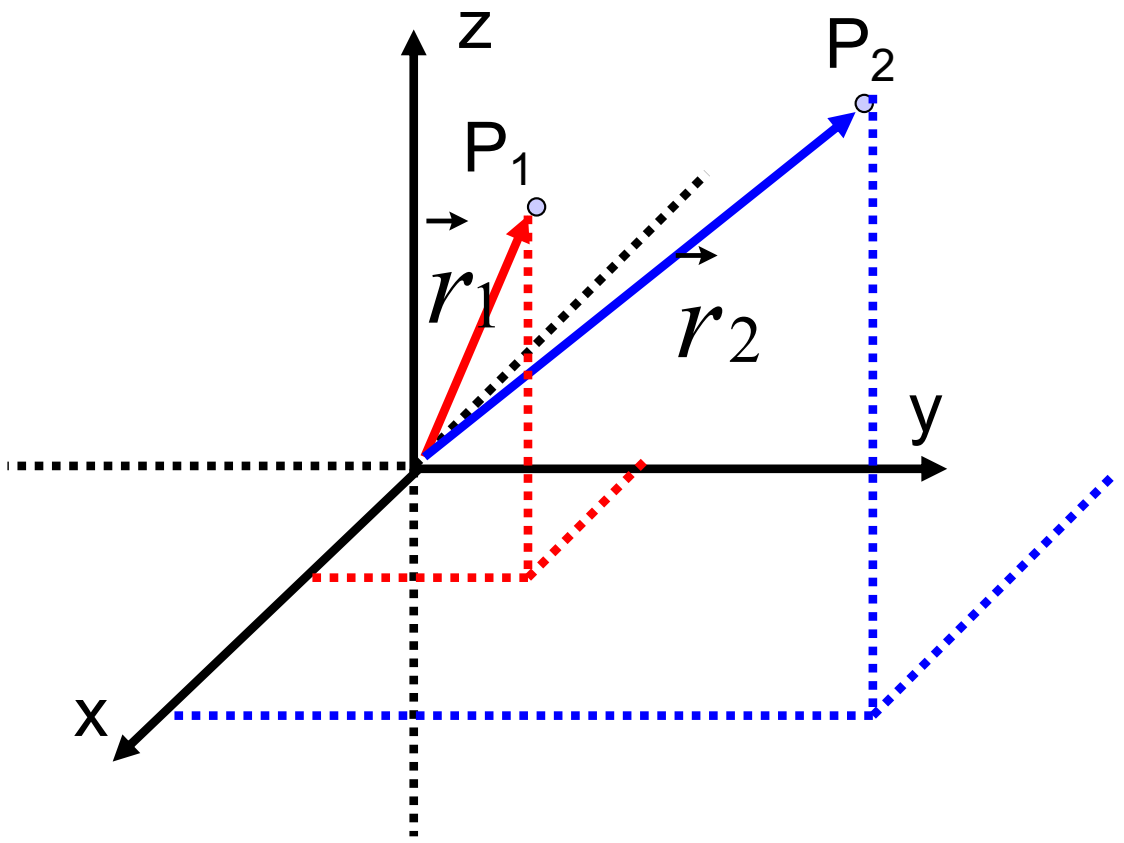
Vector From Origin to a Point



P_1: (x_1, y_1, z_1)

(Review at home)

Vector Between Two Points

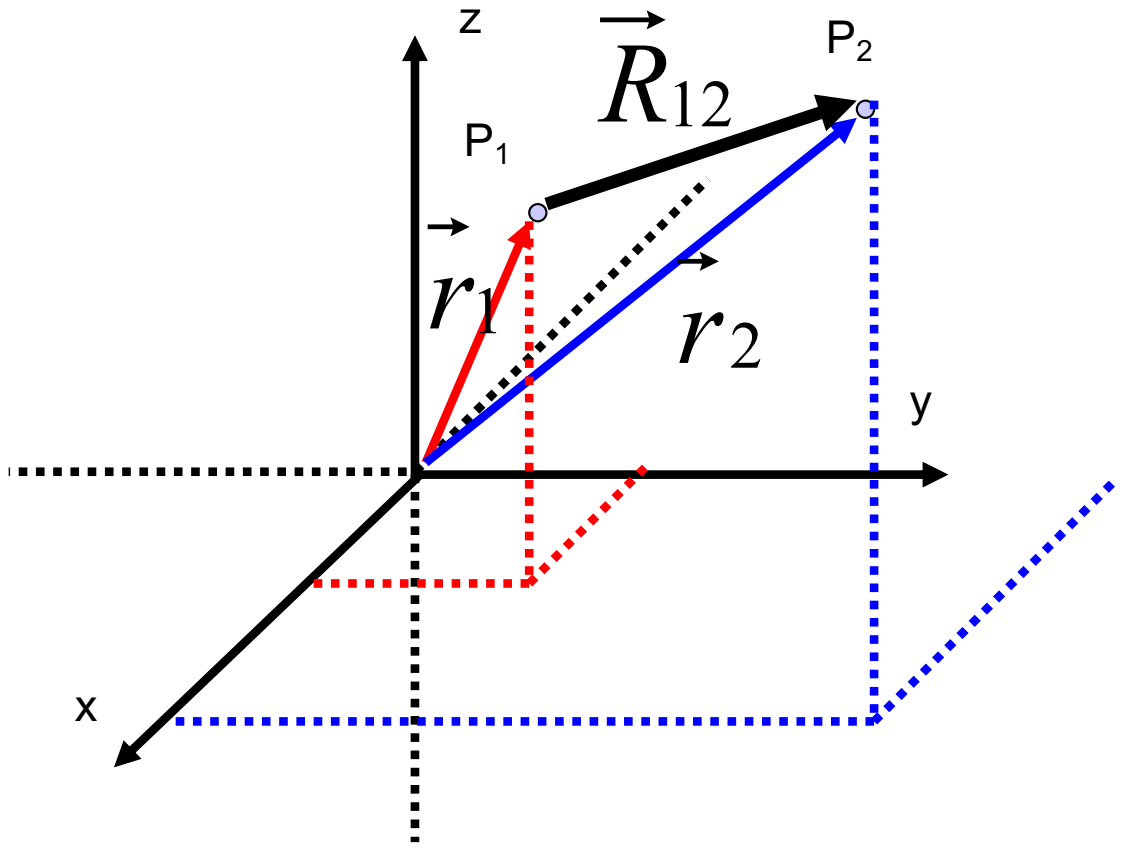


P_1: (x_1, y_1, z_1)

P_2: (x_2, y_2, z_2)

(Review at home)

Vector Between Two Points



$$\vec{R}_{12} = (FinalPosition) - (InitialPosition)$$

$$\vec{R}_{12} = (x_2 - x_1)\hat{a}_x + (y_2 - y_1)\hat{a}_y + (z_2 - z_1)\hat{a}_z$$

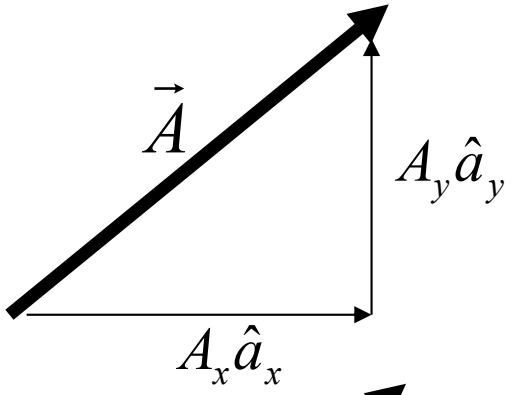
(Review at home)
Vector Math

$$\vec{A} = A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z$$

$$\vec{B} = B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z$$

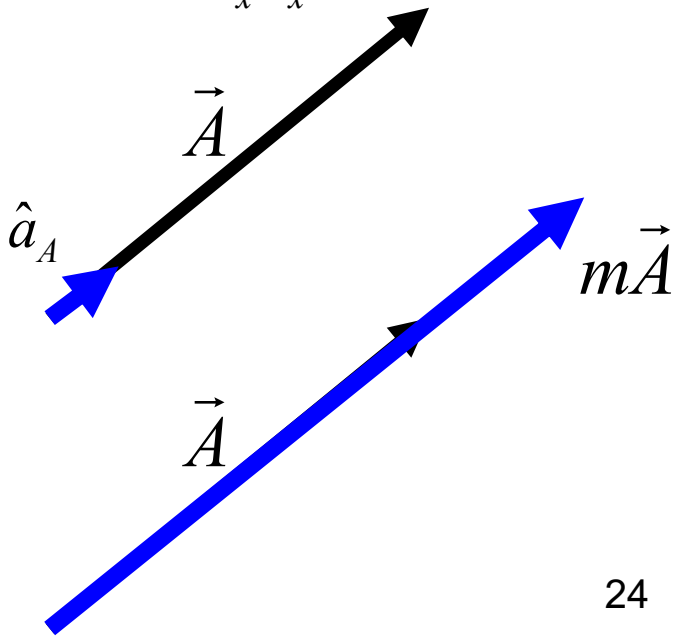
Magnitude of **A**

$$\left|\vec{A}\right| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$



Unit vector in direction of **A**

$$\hat{a}_A = \frac{\vec{A}}{\left|\vec{A}\right|} = \frac{A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z}{\sqrt{A_x^2 + A_y^2 + A_z^2}}$$

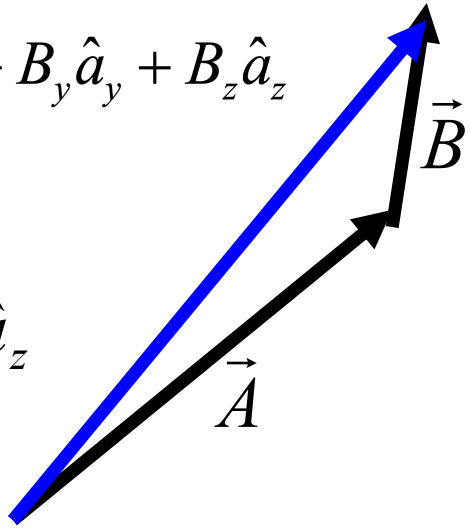


Multiplication of **A** by a scalar

$$m\vec{A} = mA_x\hat{a}_x + mA_y\hat{a}_y + mA_z\hat{a}_z$$

(Review at home)
Vector Math

$$\vec{A} = A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z \qquad \vec{B} = B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z$$

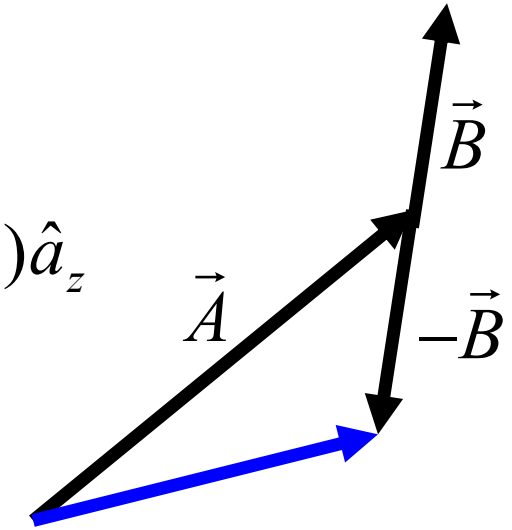


Addition of **A** and **B**

$$\vec{A} + \vec{B} = (A_x + B_x)\hat{a}_x + (A_y + B_y)\hat{a}_y + (A_z + B_z)\hat{a}_z$$

Subtraction of **A** and **B**

$$\vec{A} - \vec{B} = (A_x - B_x)\hat{a}_x + (A_y - B_y)\hat{a}_y + (A_z - B_z)\hat{a}_z$$

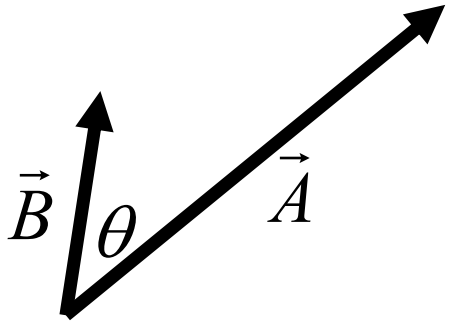


(Review at home)
Dot Product

$$\vec{A} = A_x \hat{a}_x + A_y \hat{a}_y + A_z \hat{a}_z \qquad \vec{B} = B_x \hat{a}_x + B_y \hat{a}_y + B_z \hat{a}_z$$

Dot Product of **A** and **B**

$$\vec{A} \bullet \vec{B} = \left|\vec{A}\right|\left|\vec{B}\right| \cos\theta$$



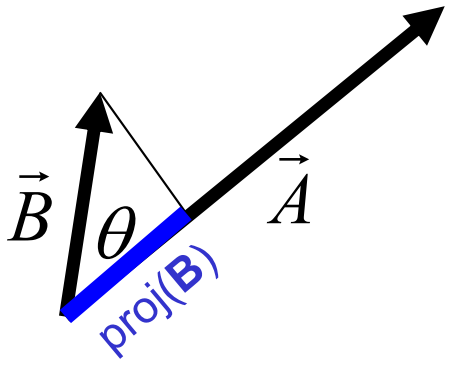
Resulting quantity is a SCALAR

Physical Meaning

(Magnitude of **A**)*(Projection of **B** onto **A**)

or (Magnitude of **B**)*(Projection of **A** onto **B**)

$$\vec{A} \bullet \vec{B} = \vec{B} \bullet \vec{A}$$



(Review at home)

Dot Product

$$\vec{A} = A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z \qquad \vec{B} = B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z$$

Dot Product of **A** and **B**

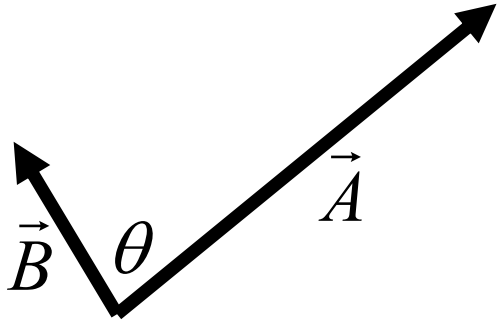
$$\vec{A} \bullet \vec{B} = \left|\vec{A}\right|\left|\vec{B}\right|\cos\theta$$

$$\vec{A} \perp \vec{B}$$

$$\cos\theta = 0$$

$$\vec{A} \bullet \vec{B} = 0$$

$$\vec{A} \perp \vec{B} \Leftrightarrow \vec{A} \bullet \vec{B} = 0$$

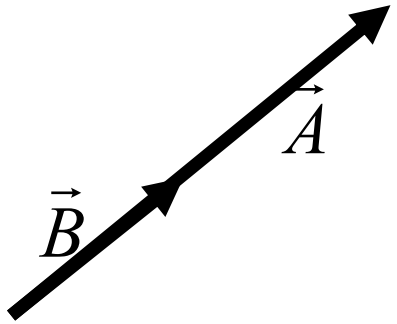


$$\vec{A} \| \vec{B}$$

$$\cos\theta = 1$$

$$\vec{A} \bullet \vec{B} = \left|\vec{A}\right|\left|\vec{B}\right|$$

$$\vec{A} \| \vec{B} \Leftrightarrow \vec{A} \bullet \vec{B} = \left|\vec{A}\right|\left|\vec{B}\right|$$



(Review at home)

Dot Product

$$\vec{A} = A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z \qquad \vec{B} = B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z$$

$$\vec{A} \perp \vec{B} \Leftrightarrow \vec{A} \bullet \vec{B} = 0$$

$$\vec{A} \parallel \vec{B} \Leftrightarrow \vec{A} \bullet \vec{B} = \left|\vec{A}\right|\left|\vec{B}\right|$$

Dot Product of **Unit Vectors**

$$\hat{a}_x \bullet \hat{a}_x = 1 \quad \hat{a}_y \bullet \hat{a}_x = 0 \quad \hat{a}_z \bullet \hat{a}_x = 0$$
$$\hat{a}_x \bullet \hat{a}_y = 0 \quad \hat{a}_y \bullet \hat{a}_y = 1 \quad \hat{a}_z \bullet \hat{a}_y = 0$$
$$\hat{a}_x \bullet \hat{a}_z = 0 \quad \hat{a}_y \bullet \hat{a}_z = 0 \quad \hat{a}_z \bullet \hat{a}_z = 1$$

$$\vec{A} \bullet \vec{B} = \left(A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z\right) \bullet \left(B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z\right)$$

$$\vec{A} \bullet \vec{B} = A_xB_x + A_yB_y + A_zB_z$$

Scalar

Discussion Problem

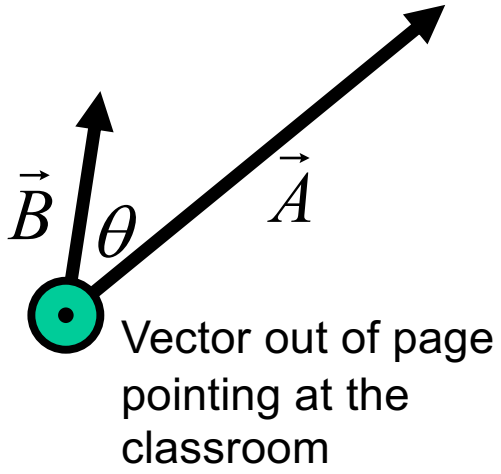
- Given **A**=$3\hat{x}+2\hat{y}+\hat{z}$, **B**= $\hat{x}+\hat{y}-\hat{z}$, **C**= $\hat{x}+2\hat{y}+3\hat{z}$, find:
  a. |**A**+**B**-4**C**|
  b. unit vector along (**A**+2**B**-**C**)
  c. **A**·**C**
  d. is **A**·**C** orthogonal?

Cross Product

$$\vec{A} = A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z \qquad \vec{B} = B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z$$

Cross Product of **A** and **B**

Resulting quantity is a VECTOR



Magnitude

$$\vec{A} \times \vec{B} = \left|\vec{A}\right|\left|\vec{B}\right|\sin\theta$$

Direction

Perpendicular to BOTH **A** and **B**

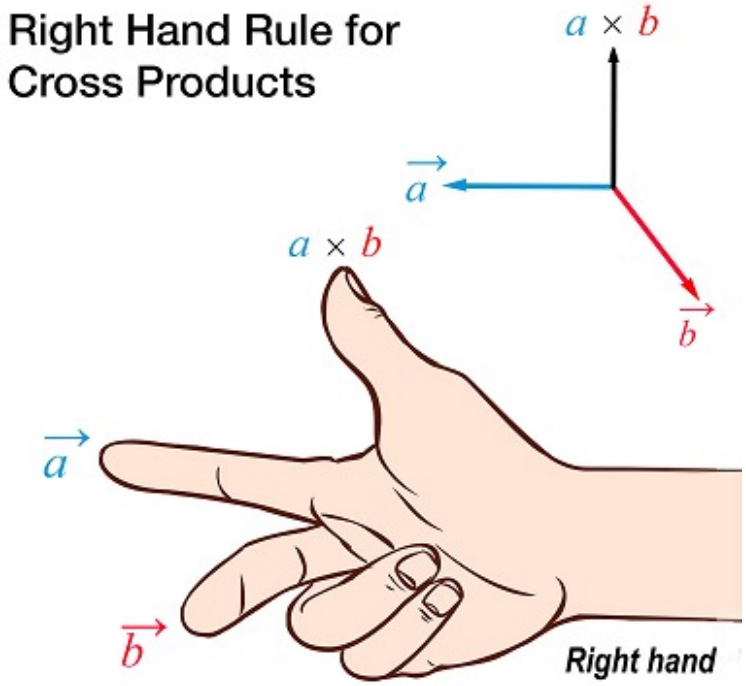
Two possible vectors satisfy this condition

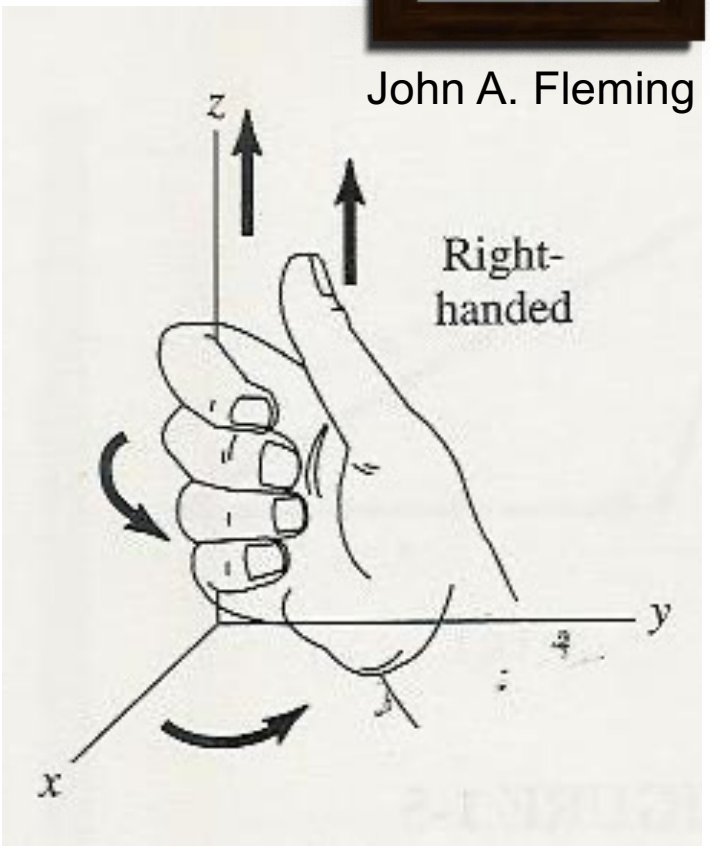
Determined using the RIGHT HAND RULE

Right Hand Rule



John A. Fleming





Cross Product

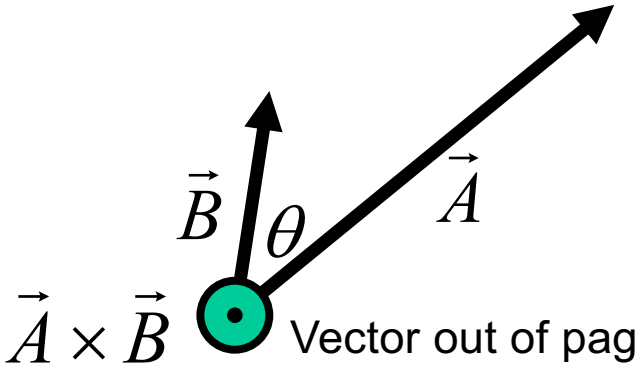
$$\vec{A} = A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z$$

$$\vec{B} = B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z$$

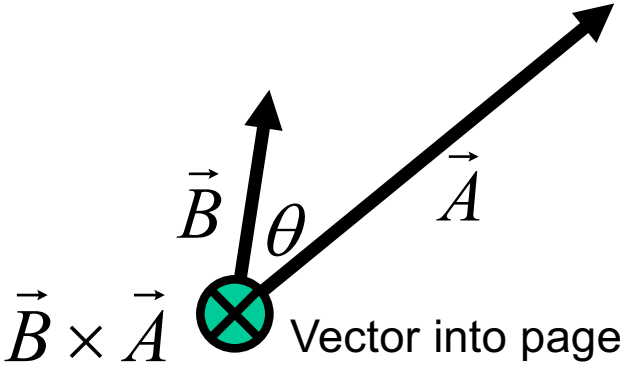
Cross Product of **A** and **B**

$$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$$

Same magnitude
Opposite direction



Vector out of page pointing at the classroom



Vector into page pointing **away** from classroom

Cross Product

$$\vec{A} = A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z \qquad \vec{B} = B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z$$

Cross Product of **A** and **B**

$$\vec{A} \times \vec{B} = \left|\vec{A}\right|\left|\vec{B}\right|\sin\theta$$

$\vec{A} \perp \vec{B}$

$$\sin\theta = 1$$

$$\vec{A}\times\vec{B} = \left|\vec{A}\right|\left|\vec{B}\right|\hat{a}_N$$

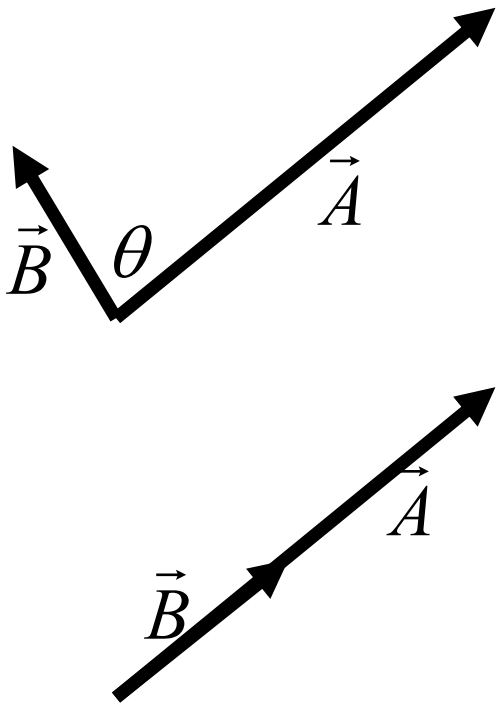
$$\vec{A} \perp \vec{B} \Leftrightarrow \vec{A}\times\vec{B} = \left|\vec{A}\right|\left|\vec{B}\right|\hat{a}_N$$

$\vec{A} \parallel \vec{B}$

$$\sin\theta = 0$$

$$\vec{A} \times \vec{B} = 0$$

$$\vec{A} \parallel \vec{B} \Leftrightarrow \vec{A}\times\vec{B} = 0$$



Cross Product

$$\vec{A} = A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z \qquad \vec{B} = B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z$$

$$\vec{A} \perp \vec{B} \Leftrightarrow \vec{A}\times\vec{B} = \left|\vec{A}\right|\left|\vec{B}\right|\hat{a}_N$$

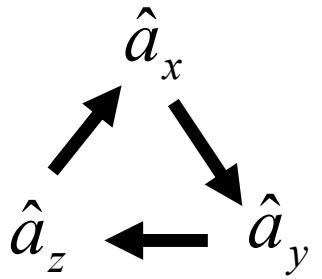
$$\vec{A} \parallel \vec{B} \Leftrightarrow \vec{A}\times\vec{B} = 0$$

Cross Product of **Unit Vectors**

$$\hat{a}_x\times\hat{a}_x = 0 \qquad \hat{a}_y\times\hat{a}_x = -\hat{a}_z \qquad \hat{a}_z\times\hat{a}_x = \hat{a}_y$$

$$\hat{a}_x\times\hat{a}_y = \hat{a}_z \qquad \hat{a}_y\times\hat{a}_y = 0 \qquad \hat{a}_z\times\hat{a}_y = -\hat{a}_x$$

$$\hat{a}_x\times\hat{a}_z = -\hat{a}_y \qquad \hat{a}_y\times\hat{a}_z = \hat{a}_x \qquad \hat{a}_z\times\hat{a}_z = 0$$

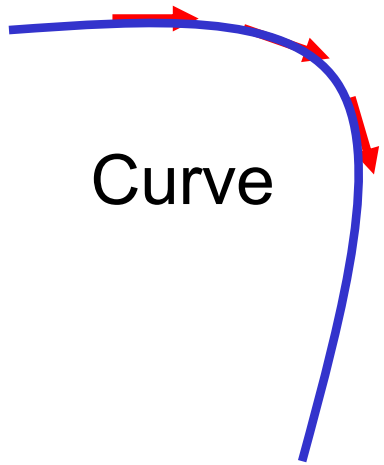


RH Coordinate System
Sign Convention

$$\vec{A}\times\vec{B} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = \underbrace{(A_yB_z - A_zB_y)\hat{a}_x + (A_zB_x - A_xB_z)\hat{a}_y + (A_xB_y - A_yB_x)\hat{a}_z}$$

Vector
Perpendicular to **A** & **B**
Right Hand Rule to Select Direction

Differential Length Vector



Always tangent to a curve or surface

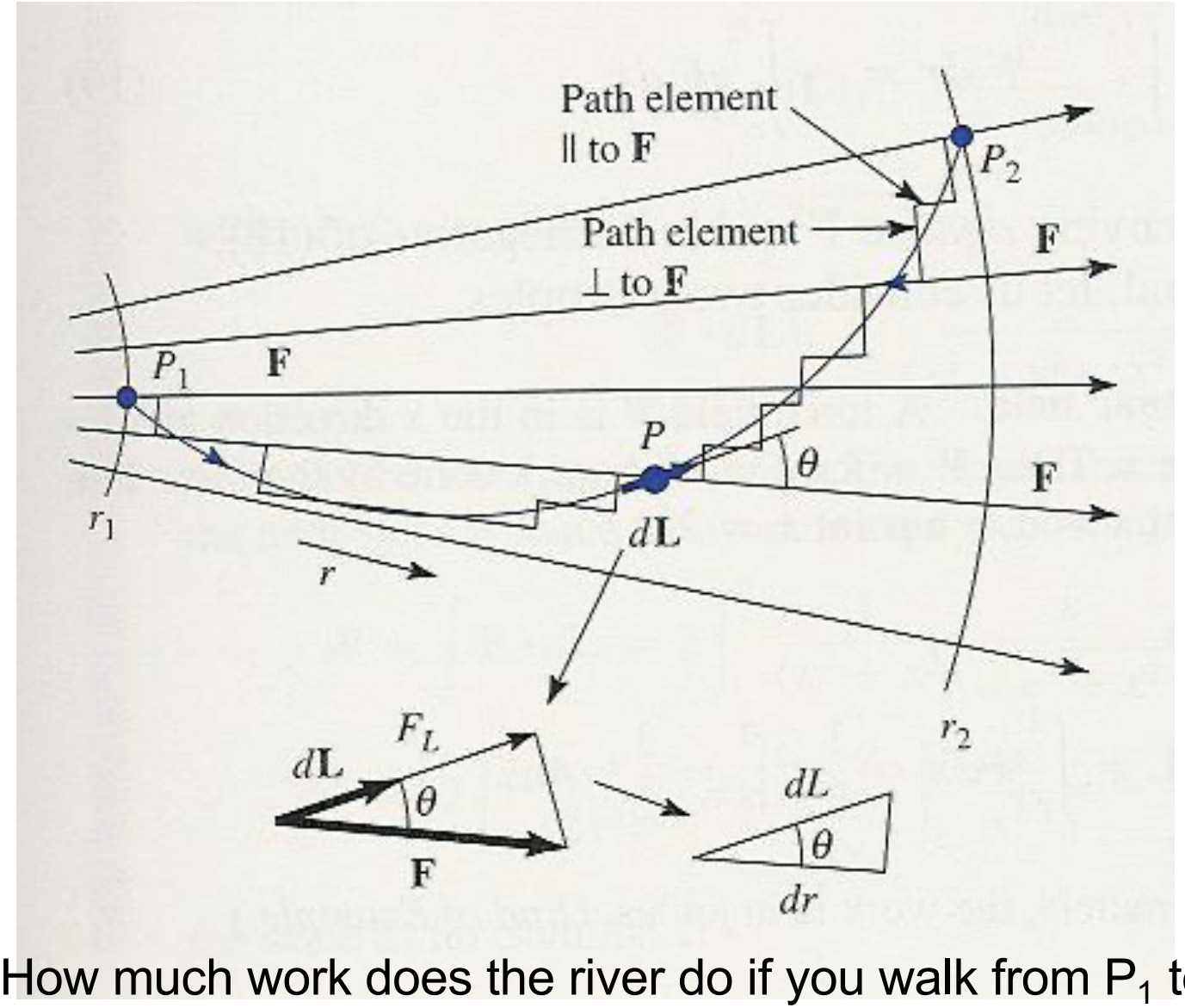
Exact vector is different at different places on the curve or surface

What we will plug into Faraday's law

$$d\vec{l} = dx\hat{a}_x + dy\hat{a}_y + dz\hat{a}_z$$

(in cartesian coordinates only)

Imagine you are *walking* in a river and the water pushes with a force, **F**, determined by the river's VELOCITY *VECTOR FIELD*.



How much work does the river do if you walk from P_1 to P_2 along the path shown?

Discussion Problem

- Given **A**=$3\hat{x}+2\hat{y}+\hat{z}$, **B**= $\hat{x}+\hat{y}-\hat{z}$, **C**= $\hat{x}+2\hat{y}+3\hat{z}$, find:
  a. **B** x **C**
  b. **A** $\cdot$ (**B** x **C**)

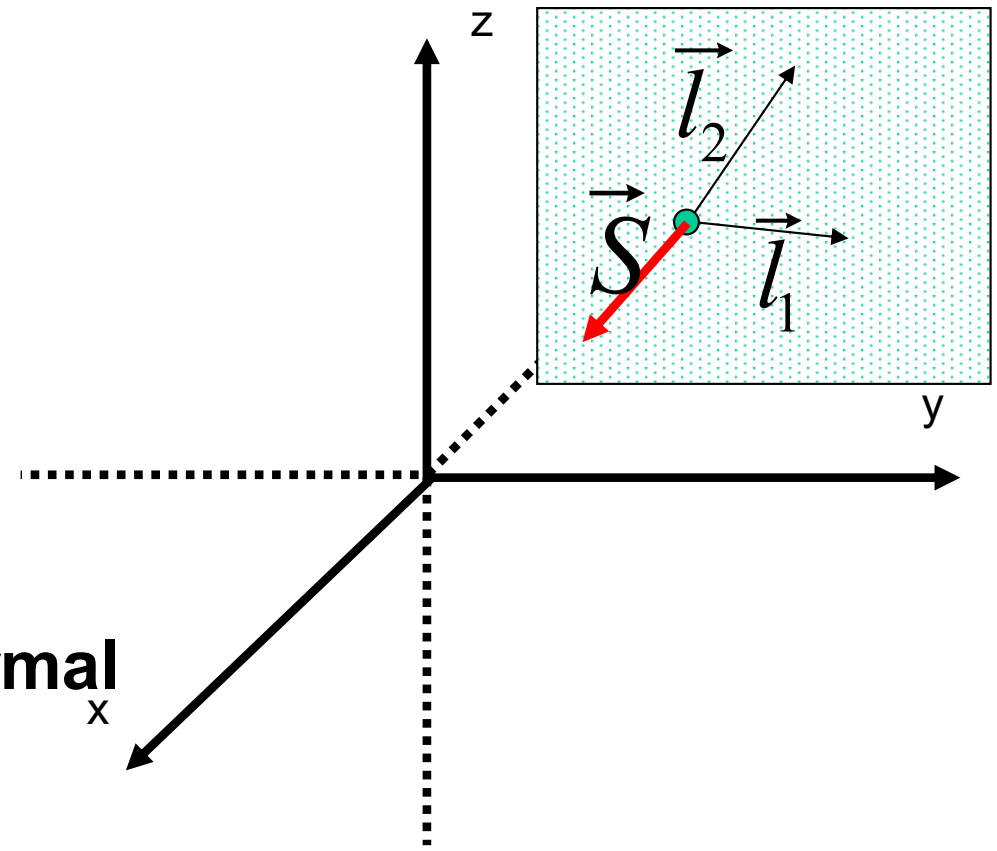
Surface Vector

Easy example surface:
Flat surface in the yz plane

Pick a point on the surface

Find two vectors at that point that are tangent to the surface

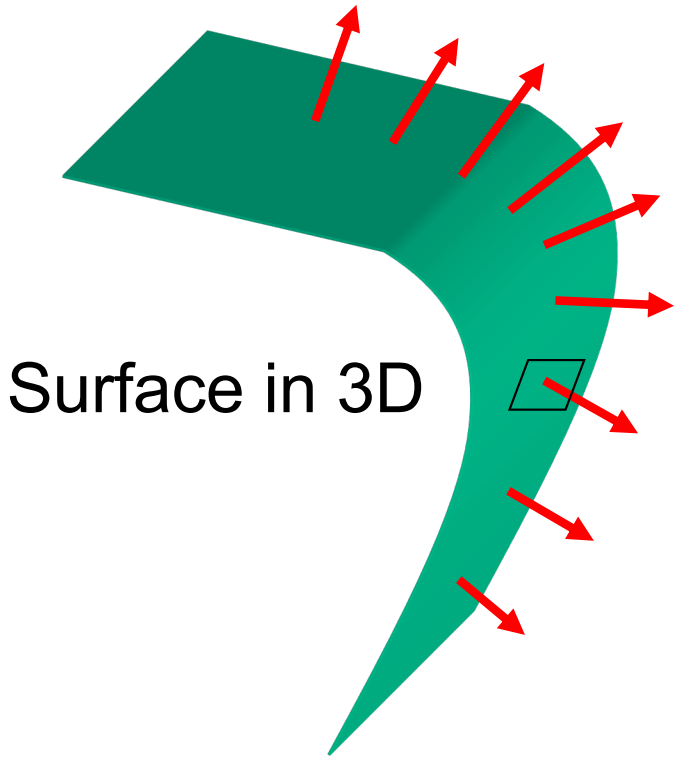
A **surface vector** is always **normal** (perpendicular) to the surface



You get **normal** vectors by performing a **cross product**

$$\vec{S} = \vec{l}_ \times \vec{l}_$$

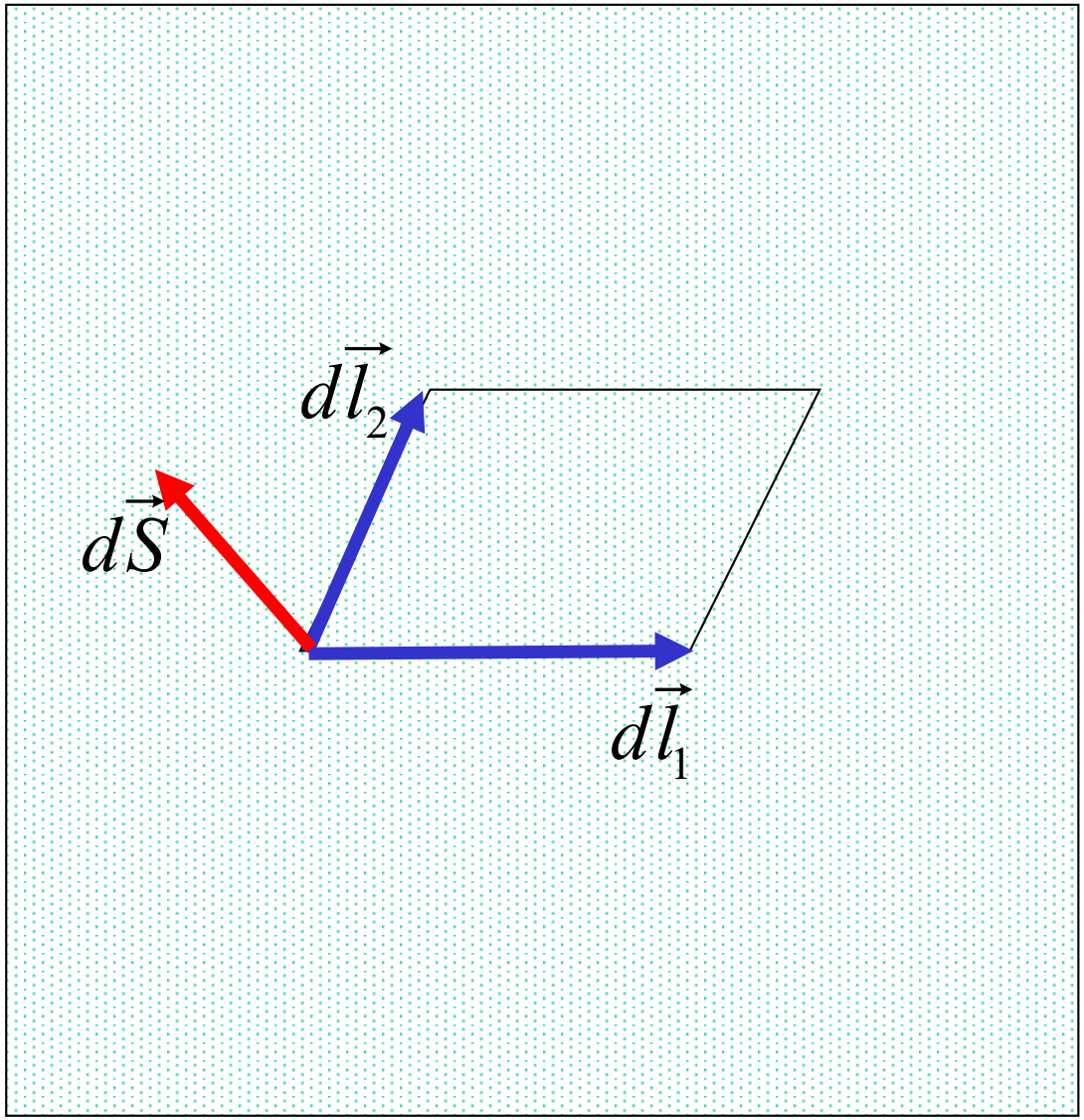
Surface Vectors



How do we come up with a surface vector when a surface is curved?

If we zoom into a small enough area, the surface will “look” flat

Differential Surface Vectors



Find two differential length vectors at the point on the surface.

Think of d**S** as a tiny parallelogram with sides bounded by d$\mathbf{l}_1$ and d$\mathbf{l}_2$ and direction perp to the surface.

$$d\vec{S} = d\vec{l}_1 \times d\vec{l}_2$$

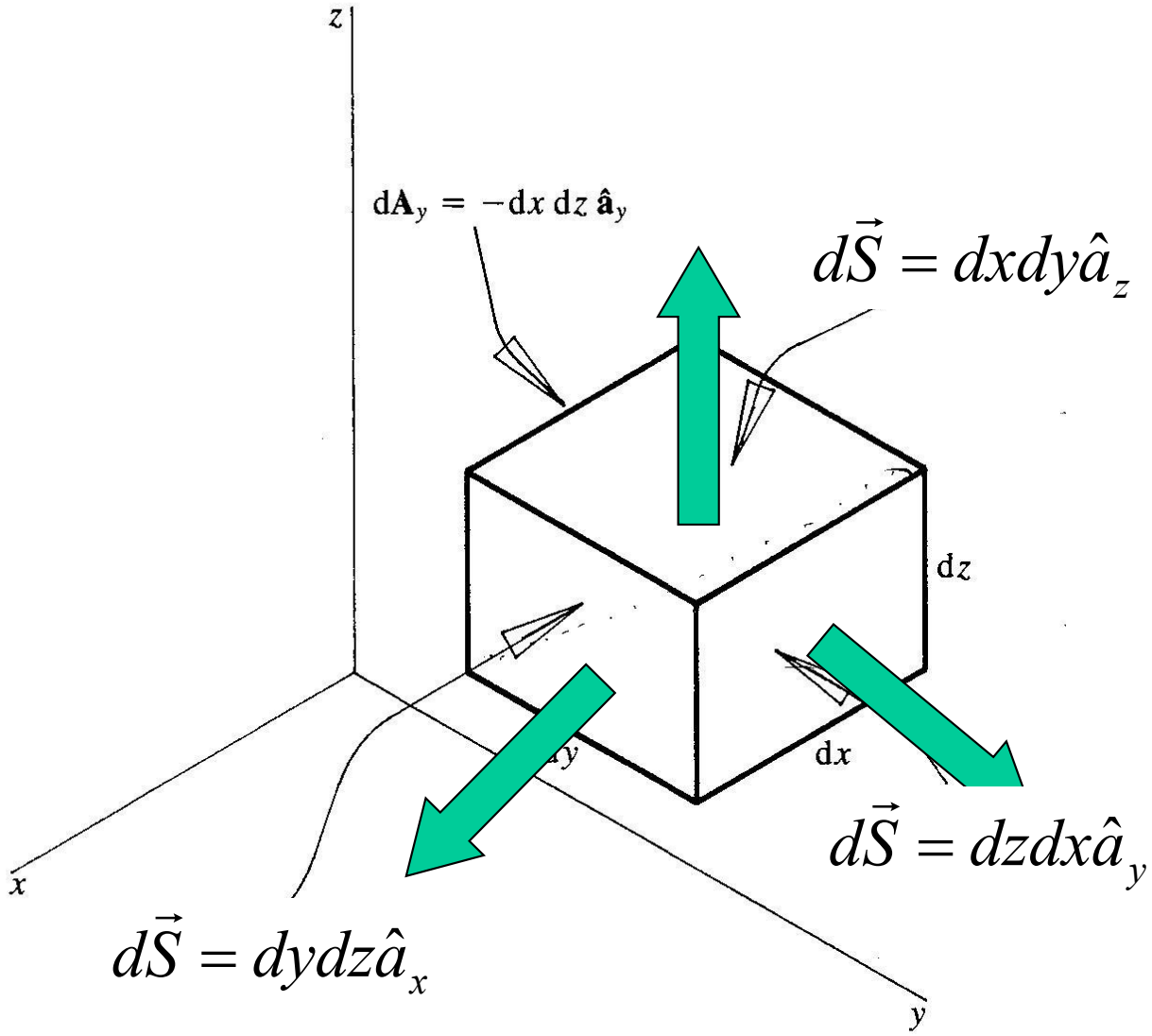
Cartesian Coordinates

(x,y,z)

$$d\vec{l} = dx\hat{a}_x + dy\hat{a}_y + dz\hat{a}_z$$

$$d\vec{S} = \pm dydz\hat{a}_x$$
$$d\vec{S} = \pm dzdx\hat{a}_y$$
$$d\vec{S} = \pm dxdy\hat{a}_z$$

Differential area or surface vectors for unit vectors in cartesian coords



Lecture 1 Summary

- Dot Product $\vec{A} \bullet \vec{B} =$ _____ = _____
- Cross Product

$$\vec{A} \times \vec{B} = \begin{vmatrix} _ & _ & _ \\ _ & _ & _ \\ _ & _ & _ \end{vmatrix} = <_____, _____, _____>$$

- Differential vector

$$d\vec{l} = dx\hat{a}_x + dy\hat{a}_y + dz\hat{a}_z$$

$$d\vec{S} = \pm dydz\hat{a}_x$$
$$d\vec{S} = \pm dzdx\hat{a}_y$$
$$d\vec{S} = \pm dxdy\hat{a}_z$$

Next

- Next two lectures:
  - Scalar and Vector Fields
  - The Lorentz Force
  - Coulomb's Law
  - Surf. Integrals/Gauss' Law